

## Task 3b - Evaluation

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### *Greenfield Local Hub Web Application*

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### Writing Plan

Before writing the evaluation, I looked back over the Task 1 proposal, the design documents and the test strategy, then compared them against the Greenfield Local Hub application which was actually developed. I wanted to make sure this evaluation was based on the prototype and not just a generic reflection, because the main point is to see if what I said I would make for Greenfield Local Hub was actually made in the final web application.

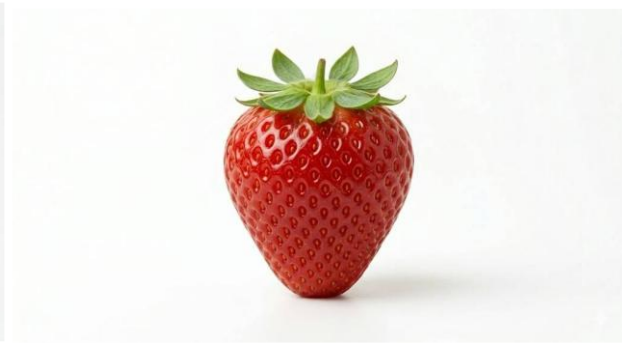
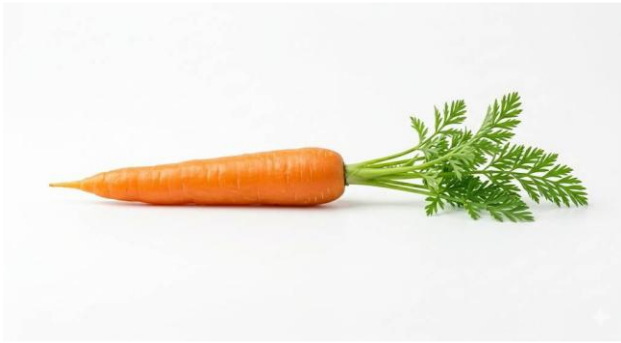
- Firstly, I would look at the assets and content used, like the producer images, product images, home page images and the written information shown to users.
- After this I would compare the original requirements with what was developed, such as accounts, products, producers, basket, checkout, order tracking, accessibility and dashboards.
- I would then look at the non-functional requirements and KPI, but I would also be honest where there is not full evidence, for example page speed graphs or final user feedback graphs.
- Lastly, I would explain future improvements which would make the solution better for customers, producers and Greenfield Local Hub as a business.

### Assets and Content

#### How appropriate were the assets used in your prototype?

The assets used in my prototype were appropriate because they matched the purpose of Greenfield Local Hub, which is based around local produce, producers and customers being able to clearly see what they are buying. For example, the application uses images of products like fruit, vegetables, dairy and meat, as well as producer images and home page images which make the website feel more suited to a local food business. This is important because the target audience is likely to be local households and families who want to support local producers, so the images help them quickly understand the products and the values of the business.

## Greenfield Local Hub Solution Evaluation



In my proposal I said that the website should use a green and white style because this links to nature, sustainability and local produce. This was carried through into the actual Greenfield Local Hub application as the pages use a clean green style with product cards and producer sections. Because of this, the assets were not just there to make the pages look nicer, they also supported the purpose of the website by making it clearer and more trustworthy for users who may not be very technical.

### **How valid and reliable were the sources used?**

The sources I used in Task 1 were Seasonal Roots and The Produce Box. These were useful and valid sources because they are real produce/co-operative style websites and they showed common conventions that I could compare against, such as having an account system, product browsing, producer information and a clear modern layout. This had a significant impact on my web application because the application ended up having similar features like the product listings, producer pages, account login, basket and checkout system.

These sources were reliable for looking at design conventions, however they were not perfect sources for everything because they were competitor websites rather than academic or official sources. This was still suitable because I was researching what an industry standard produce website should feel like. If I was improving the project, I would create a clearer asset log showing where each image came from, what licence it had and whether it needed crediting, because this would make the content evidence stronger.

### **Legal and ethical implications**

There were legal and ethical implications when using assets, mainly around copyright and whether the images are allowed to be used. The product and producer images should either be original, open source, licensed or credited if needed. If this was not checked properly then it could cause copyright issues for Greenfield Local Hub, so this is something which would need to be handled carefully before the website was used for a real business.

There were also ethical issues around making sure the product information is honest. In the application for Greenfield Local Hub, products include information such as price, availability, stock and allergens, which is important because customers should not be misled about what they are buying. Not only this, accessibility was also an ethical and legal consideration because the proposal said the solution should support users with different needs. The actual application includes font size changes, high contrast, dark mode and read aloud features, which helps make the website more inclusive.

## **Proposed Design and Final Front End Comparison**

In the design proposal I made, alongside the wireframes which were made, it was planned for the front end to primarily make use of a clean green with a white styling, because this matched the local producer and co-operative theme of Greenfield Local Hub. The wireframes included a forest green navigation bar and footer, hero sections, products cards, producer pages and a basket layout with dashboard pages. The final front end design still follows this idea because the actual solution made for Greenfield Local Hub uses the same sort of natural green style, with very clear navigation and imagery for product/producers, which makes it feel very suitable for the target audience, although now more modern and inviting.



The Products page and the Producer pages are very close to the design which was proposed, as they still make usage of product images, cards, producer information, prices, stock/availability details and clear sections for the user to read through. This is better than a plain table sort of layout as customers can just scan the page quickly which allows for more informed decisions to be made. Because of this, the final front end keeps the main purpose of the wireframes, which was to make local products and producers very easy to understand.

A few changes were made when developing the final design as the application had to work well with the database, controllers and roles rather than only looking exactly as the wireframe does. For example, the designs of the producer and admin dashboard instead because more focused-on filtering, tables, status buttons and action buttons for management as producers and admins need to complete tasks quickly. This was better for the solution developed since it made the dashboard a lot more practical for real world usage rather than just matching the original design visually.

Not every part of the wireframe was developed the exact same way. As an example, the full loyalty rewards account page wasn't developed the same way, and some page layouts were simplified so that the main features like checkout, basket, order tracking and dashboards work as intended. I think the changes were suitable because the most important part was the functionality of the prototype, and most changes made actually led to the website feeling more inviting, modern and a memorable experience which is clear to users. Overall, the final front end design developed still follows the planned brand style and layout ideas, but it changed slightly so that the application is able to work better as a real ASP.NET MVC C# prototype.

The feedback which was gathered in Task 3a also supported this design choice, all three of the non-technical users made use of the application and described their overall experience as excellent, and all of them said the green colour scheme fit the fresh local produce branding very well. Because of this, the feedback shows that the final front end was most successful at matching the wireframe idea, even if a few pages like order confirmation page could still have improvements.

### Project Outcomes

Overall, I think the prototype met most of the original requirements that were set out in the proposal. The original aim was to develop a user-friendly web application for Greenfield Local Hub where customers could view local producers, browse products with prices and availability, create an account, place orders for collection or delivery and track their orders. In the final solution for Greenfield Local Hub, these main parts were developed using ASP.NET MVC, Identity accounts, database backed products, baskets, orders and role-based dashboards.

The application also developed more management features than a basic customer website, because there is an admin dashboard and producer dashboard. This means producers can manage their own products and deal with producer orders, whilst admins can manage wider parts of the business. Because of this, the final outcome is suitable for the client problem as it supports both customer use and business management.

The main part that was not fully developed exactly like the proposal was the loyalty scheme. The app does include discounts, a health bundle discount and free delivery style rewards, which does support customer retention, but it is not a full points system with a proper loyalty page in the

customer account. So, this requirement was partly met, but not completely in the same way I originally planned.

Overall, these model changes made the final solution better because the app became more realistic and easier to manage. The changes supported stock control, order tracking, producer order management, contact messages, allergens and delivery/collection details. This means the final models were more detailed than the first proposal, but they were changed for a good reason because they made the application work better as a real prototype for Greenfield Local Hub.

The ContactInquiry model was also added so that messages from the Contact Us page could be saved and replied to by admins. This made the contact feature more useful because it was not just a front-end form, it became part of the admin workflow. Because of this, the model changes improved the application overall as they made the database match the actual needs of customers, producers and admins rather than only matching the first design idea.

I also added a ProducerOrders model, which was one of the most useful changes. This allowed one customer order to be split between different producers, so each producer can manage only their own part of the order. This is better for Greenfield Local Hub because the business works with multiple local producers, and it would not be practical if every producer could see or control the whole order. Not only this, it also improves data protection because producers only need access to the order items that relate to them.

Another change was improving the Orders model so that it could support delivery, collection, order status, tracking status, billing address, delivery address and terms being accepted. This was better than a simpler order model because the proposal said customers should be able to place orders for collection or delivery and view order progress. If the order model did not store this information properly, the checkout and tracking system would not have worked as well.

One important model change was adding more detail to the Products model. Instead of only storing basic product information, the final model includes stock quantity, availability, category, description, image path and allergens. This was better because customers need to know if a product is available before they order it, and allergen information is important for food products because users should be able to make safe and informed choices. Because of this, the product model became more useful for both customers and producers.

During development I made changes to the original model design because when I started building the actual Greenfield Local Hub application, some of the first planned models were not detailed enough to support the features properly. In the proposal and design documents I planned the main data around users, producers, products, baskets and orders, however when the application was developed, I had to make the models more suitable for the real workflows in the website.

## Model Changes

During development I made changes to the model design because once I started building Greenfield Local Hub, some of the first planned models were too simple for what the website actually needed to do. In the design documents I planned the main data around users, producers, products, baskets and orders, however the final application needed more detail so the customer, producer and admin workflows could work properly.

## Products Model

One change I made was improving the Products model. At first, the product data could have just stored basic information like name and price, but this would not have been enough for a real food ordering website. Because of this, the final Products model includes stock quantity, availability, category, description, image path and allergens.

This made the product model better because customers can see whether a product is actually available before ordering it, and producers can manage stock more clearly. Not only this, allergen information is important because food products need to give customers enough information to make safe choices. This links back to the proposal because I said customers should have transparent product information.

## Orders Model

Another model change was improving the Orders model. The final Orders model stores delivery, collection, delivery method, collection date, billing address, delivery address, order status, tracking status and whether terms were accepted. This was better than having a basic order model because the proposal said customers should be able to place orders for either collection or delivery and also track the progress of their order.

Because of this, the order model supports the checkout process properly instead of only saving a total price. It also means the application can validate addresses, calculate delivery costs, show order progress and keep order information linked to the correct user. This improves the project because the order system became more realistic and useful for the customer.

## ProducerOrders Model

A very important model change was adding the ProducerOrders model. This model lets one customer order be split into different producer parts. This was needed because Greenfield Local Hub is based around multiple local producers, so one customer basket could include products from more than one producer.

This change made the application better because each producer can manage only their own part of an order rather than seeing or changing the full customer order. Because of this, it supports the producer dashboard and also improves data protection, as producers only access the order items that relate to them. It also makes the business workflow better because order management is shared between producers instead of everything being handled manually by one admin.

## ContactInquiry Model

I also added the ContactInquiry model so that messages from the Contact Us page could be stored in the database and replied to by admins. This was better than having a contact form that only looked like it worked on the front end, because it made the contact system part of the actual application workflow.

This improves the project because customers can send a message and admins can manage enquiries properly. Not only this, it gives Greenfield Local Hub a clearer way to communicate with users, which supports trust and makes the web application feel more complete.

## Overall Evaluation of Model Changes

Overall, the model changes made the final solution better because the database became more suited to the real features of the application. The changes supported stock control, availability, allergens, delivery and collection details, order tracking, producer order splitting and contact

messages. This means the final models were more detailed than the first design idea, but they were changed for a good reason because they helped the prototype work better for customers, producers and admins.

## Functional Requirements

The prototype meets most of the functional requirements from the brief and proposal. This is because the key customer journey works from browsing products, viewing producers, adding to basket and placing an order. The table below shows how the main requirements compare to the final application.

| Requirement                    | Outcome    | Evaluation                                                                                                                                                                                   |
|--------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Account registration and login | Met        | The app uses ASP.NET Identity and has different roles such as Standard, Producer, Admin and Developer. This meets the proposal because users can log in and have protected account features. |
| Product information            | Met        | Products show name, price, category, image, description, stock, availability and allergens. This meets the need for transparent product information.                                         |
| Producer information           | Met        | Producer pages and records show producer names, emails, images, VAT information and descriptions, which supports the aim of showing local producer information.                              |
| Basket and checkout            | Met        | Users can add products to a basket and checkout. The system checks stock and calculates totals on the server side, which makes it more reliable.                                             |
| Collection and delivery        | Met        | Checkout supports delivery or collection, including delivery costs, address validation and collection date validation.                                                                       |
| Order tracking and management  | Met        | Orders have order status and tracking status. Admins and producers can manage statuses, and producer orders are split so producers only deal with their own parts.                           |
| Producer dashboard             | Met        | Producers can view their products, manage stock and availability, and accept or cancel producer order items.                                                                                 |
| Accessibility features         | Mostly met | The app has font size controls, high contrast, dark mode and read aloud support. This meets the main idea, but formal WCAG testing would make it stronger.                                   |
| Loyalty scheme                 | Partly met | Discounts and free delivery rewards exist, but there is not a full loyalty points page like the proposal originally described.                                                               |

Because of this, the functional requirements were mostly successful. The most important thing is that the app does not only show static pages, but stores information in the database and uses real roles and controllers to manage products, orders and users. The only main functional gap is that the

loyalty scheme should be developed further so that users can properly see and understand their rewards.

The technical paired code review further supported this as the code was mostly stated as easy to understand and review, alongside no major issue standing out in most areas. This helps show that the main features were not only working for users, but were also very understandable from a technical point of view.

## Non-Functional Requirements

The non-functional requirements were also mostly met, especially involving security, usability and accessibility. The non-functional requirement of security was met very well because the application makes proper usage of ASP.NET Identity, alongside role-based authorisation and validation. This means that normal customers are unable to access the staff or producer areas, and producers are limited to only their own product and orders. This directly meets the proposals promise of customer and business data being protected.

Usability was met because the pages follow a very consistent layout, with the navigation being very clear and product cards which make the website feel much easier to understand. This is especially useful for the target audience since local customers are unlikely to be very well experienced with technology, so it needs to be ensured the interface is simple and somewhat predictable.

Accessibility was also supported well through the accessibility menu in the nav bar, where the users can change the font size throughout the website, use high contrast, dark mode and read aloud features.

Performance and scalability are much harder to fully prove as I do not have final load testing graphs or page speed evidence in the files. It's suitable as a prototype and uses normal MVC/database patterns, however the proposal included targets like pages loading under two seconds and supporting many users. These would need more testing before saying the solution is fine to be used as a real system.

The non-technical questionnaire also supported the usability and accessibility requirements because all three of the users said that the main menu was very easy to understand and that the checkout fields were very clear, and also that adding a product to the basket was very easy to do. This makes the non-functional requirements much stronger because it shows that the design was not only planned for usability, but the users also responded to it positively.

## KPI

The prototype meets several of the KPI from the proposal. For example, users can register and log in, products and producers are loaded from the database, orders can be placed, basket totals are calculated and users receive confirmation/status information. These are important KPI because they show that the core purpose of the website works and that users can complete the main tasks without needing staff help.

The accessibility KPI is partly met because the controls are available in the application and can be accessed quickly from the interface. The security KPI is also mostly met because passwords are handled by Identity and roles are used to protect dashboards. However, some KPI such as load times, supporting 1000 active users and product search under heavy load are not fully proven



because there is no final performance evidence included. The impact of this is that the application is good as a prototype, but more testing would be needed before Greenfield Local Hub used it as a real system.

## User Acceptance Criteria

The prototype meets most of the user acceptance criteria from the proposal. For a new customer, the user can create an account, log in, browse products, view product details, add items to a basket and place an order. This meets the main customer need because the user can actually go through the buying process rather than only reading information.

For a returning customer, the app supports account-based order features and checkout rewards like discounts and free delivery rewards. This helps with the original aim of encouraging customers to return, although it would be better if there was a clearer loyalty points page. For local producers, the producer dashboard meets the requirement because producers can manage their own products and respond to their own order items. This supports the business because it reduces the need for all order and stock management to be done manually by staff.

The accessibility acceptance criteria are mostly met because the user can adjust the appearance of the website. The main limitation is that I cannot prove every interactive element has been tested with keyboard navigation or WCAG checks, so this would be a future improvement. Overall, the user acceptance criteria were met well enough for a prototype, with the loyalty and formal accessibility testing being the main things to improve.

## Feedback Gathered

The feedback for the application was gathered using a technical paired coding review, a non-technical questionnaire and an observation sheet. The technical review was useful because it went over the codebase, controllers, roles, validation, dashboards and the overall maintainability of the solution developed. On the other hand, the technical questionnaire was useful because it focused more on the user experience, navigation, accessibility, colours, product pages, basket and checkout.

Overall, the feedback which was gathered was overall positive. In the non-technical questionnaire, all three of the users who did the questionnaire rated the overall experience as excellent and all three said the menu was very easy to understand, and they all also completed a delivery order. This shows the main customer journey was successful because users are able to browse, add to basket and go through the checkout system without any major confusions.

The feedback provided also showed that some of the features in the application were especially strong. Users had mentioned the home design, product pages, shop filters, basket, contact page and accessibility tools as their favorite parts of the experience. The technical review also showed that most areas in the application were easy to understand and review which further supports the idea that the project was developed in a way which is clear enough for an individual to inspect.

However, the feedback also gave very useful improvements. Some users said that the order confirmation page could have a visual redesign, the growers page had a stutter on one device, and the read aloud feature would be better if it was able to read highlighted text or text near where the user clicks. Because of this, the feedback overall shows the prototype was successful, but the next version should focus on small usability features alongside further accessibility implantation and some improvements for the performance of the solution.

## Future Improvements

The main future improvement would be to finish the loyalty scheme properly. This would include a customer account page where users can see loyalty points, rewards they have earned, discounts available and how close they are to their next reward. This would improve the project because it would make the reward system clearer and would give customers more reason to keep using Greenfield Local Hub.

The feedback also supports making improvements carefully instead of just changing everything, users were mostly positive about the current design and functionality. Because of this, future changes focus on the area's users mentioned the most like the order confirmation page not matching the design of other pages well, clearer menus, accessibility improvements and small performance fixes like supposed stuttering on the grower's page.

Another improvement would be collecting more feedback and adding graphs from this feedback. For example, users could be asked how easy the basket was to use, if the product information was clear, if the accessibility controls helped them and if the checkout process was simple. This would make the evaluation stronger because it would be supported by real evidence rather than only comparing the finished app to the proposal.

The project could also be improved by adding automated emails for order updates, live payment integration, stronger image source documentation, more detailed producer analytics and proper performance testing. These improvements would help customers by making orders clearer, help producers by making management easier, and help Greenfield Local Hub because the system would be more reliable and professional if it was developed past prototype stage.

## Conclusion

In conclusion, I think the solution developed for Greenfield Local Hub is a suitable prototype for the Greenfield Local Hub solution because it meets most of what I said I would create in Task 1. The strongest areas are the product browsing, producer information, basket, checkout, order management, role-based dashboards and accessibility features. The project is not perfect because the loyalty system is not fully developed into a points page and some KPI still need proper testing evidence, but overall, the prototype does meet the main purpose of helping customers buy local produce and helping producers manage their products and orders.